

Understanding the Document Object Model (DOM) in Mozilla



This article is an introduction to the DOM programming interface and the DOM inspector, which is a tool that can be used to inspect and edit the live DOM of any Web document or XUL application.

The Document Object Model (DOM) is a programming interface for HTML and XML documents. It provides a structured representation of a document and it defines a way that the structure can be accessed from the programs so that they can change the document structure, style and content. The DOM provides a representation of the document as a structured group of nodes and objects that have properties and methods. Essentially, it connects Web pages to scripts or programming languages.

A Web page is a document that can either be displayed in the browser window or as an HTML source that is in the same document. The DOM provides another way to represent, store and manipulate that same document. In simple terms, we can say that the DOM is a fully object-oriented representation of a Web page, which can be modified by any scripting language.

The W3C DOM standard forms the basis of the DOM implementation in most modern browsers. Many browsers offer extensions beyond the W3C standard.

All the properties, methods and events available for manipulating and creating the Web pages are organised into

objects. For example, the document object that represents the document itself, the `tableObject` that implements the special `HTMLTableElement` DOM interface to access the HTML tables, and so forth.

Why is DOM important?

'Dynamic HTML' (DHTML) is a term used by some vendors to describe the combination of HTML, style sheets and scripts that allow documents to be animated. The W3C DOM working group is aiming to make sure interoperable and language-neutral solutions are agreed upon.

As Mozilla claims the title of 'Web Application Platform', support for the DOM is one of the most requested features; in fact, it is a necessity if Mozilla wants to be a viable alternative to the other browsers. The user interface of Mozilla (also Firefox and Thunderbird) is built using XUL and the DOM to manipulate its own user interface.

How do I access the DOM?

You don't have to do anything special to begin using the